

Homework 4

● Graded

Student

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Total Points

32 / 32 pts

Question 1

Problem 1

3 / 3 pts

✓ - 0 pts Correct

- 1 pt incorrect set up of the integral: $\int_1^{e^2} \int_1^{e^2} \frac{\ln(x) \ln(y)}{4xy} dx dy$

- 1 pt incorrect evaluation of the integral

- 1 pt incorrect final answer: 1

- 1 pt incorrect substitution: the integral should be $\int_0^2 \int_0^2 \frac{uv}{4} du dv$

Question 2

Problem 2

6 / 6 pts

✓ - 0 pts Correct

- 0.5 pts part b: algebraic error

- 1 pt space of cdf not defined/not defined correctly

- 2 pts part a work not shown

- 2 pts errors in finding cdf

- 2 pts error in finding expected value

- 3 pts part a missing

- 3 pts part b missing

- 1 pt found E[X] but not E[2X-1]

✓ - 0 pts Correct

- 0.4 pts 3(a) $x < a$

- 0.4 pts 3(a) $a \leq x < b$

- 0.4 pts 3(a) $b \leq x < c$

- 0.4 pts 3(a) $c \leq x < d$

- 0.4 pts 3(a) $d \leq x$

3(b)

- 0.5 pts E(X) formula

- 0.5 pts E(X) answer

- 0.5 pts $E(X^2)$ formula

- 0.5 pts $E(X^2)$ answer

- 0.5 pts Var(X) formula

- 0.5 pts Var(X) answer

- 1 pt 3(b) Var(X) definition

- 0.5 pts 3(b) antiderivative

- 0.5 pts 3(b) Var(X) answer

- 5 pts Incorrect

Question 4

Problem 4

5 / 5 pts

✓ - 0 pts Correct

- 0.5 pts (a) Possible values of X : $0, 1, 2, \dots, n$
- 0.5 pts (a) formula
- 0.5 pts (a) $P(X=k \mid \text{coin1})$
- 0.5 pts (a) $P(\text{coin1})$
- 0.5 pts (a) $P(X=k \mid \text{coin2})$
- 0.5 pts (a) $P(\text{coin2})$

(b)

- 1 pt (b) reason: $P(X=k) = nCk p_1^k (1 - p_1)^{n-k}$, or valid explanation
- 1 pt (b) answer: $X \sim \text{Bin}(n, p_1 = p_2)$
- 0.5 pts (b) answer: wrong(or not mentioning) parameters $(n, p_1 = p_2)$
- 0.5 pts (b) Define what p is

- 5 pts Incorrect

Question 5

Problem 5

5 / 5 pts

✓ - 0 pts Correct

- 1 pt distribution of X incorrect
- 1 pt distribution of Y incorrect
- 3 pts parameters of distributions not explicitly given
- 5 pts missing
- 0.5 pts minor error in parameterizing of X

✓ - 0 pts Correct

- 1 pt (6-1) probability of giving a picture out on the first day is pq
- 1 pt (6-2) Let X be the number of pictures that she gives out in n days, then $X \sim Binom(n, pq)$. Then $P(X = k) = \binom{n}{k}(pq)^k(1 - pq)^{n-k}$: incorrect probability
- 1 pt (6-2) Let X be the number of pictures that she gives out in n days, then $X \sim Binom(n, pq)$. Then $P(X = k) = \binom{n}{k}(pq)^k(1 - pq)^{n-k}$: incorrect pmf
- 1 pt (6-3) $P(\text{staying home} | \text{no picture given}) = (P(\text{no picture given and staying at home})) / (P(\text{no picture given})) = (1-p) / ((1-p) + (p(1-q))) = (1-p) / (1-pq)$: incorrect formulation
- 1 pt (6-3) $P(\text{staying home} | \text{no picture given}) = (P(\text{no picture given and staying at home})) / (P(\text{no picture given})) = (1-p) / ((1-p) + (p(1-q))) = (1-p) / (1-pq)$: incorrect probability computation
- 1 pt (6-4) $P(\text{went out but did not give picture}) = p(1-q)$. Let X be days that Sarah went out but did not give out pictures, then $X \sim Binom(7, p(1 - q))$. Thus, $P(X = 0) = \binom{7}{0}(p(1 - q))^0(1 - (p(1 - q)))^7 = (1 - p + pq)^7$: incorrect probability
- 2 pts (6-4) $P(\text{went out but did not give picture}) = p(1-q)$. Let X be days that Sarah went out but did not give out pictures, then $X \sim Binom(7, p(1 - q))$. Thus, $P(X = 0) = \binom{7}{0}(p(1 - q))^0(1 - (p(1 - q)))^7 = (1 - p + pq)^7$: incorrect formulation
- 8 pts missing problem 6 solution

$$1. \quad V = \int_1^{e^2} \int_1^{e^2} \frac{\ln x \ln y}{4xy} dx dy$$

$$= \int_1^{e^2} \frac{\ln x}{4x} dx \int_1^{e^2} \frac{\ln y}{4y} dy$$

$$= \frac{1}{4} \int_1^{e^2} \frac{\ln x}{x} dx \int_1^{e^2} \frac{\ln y}{y} dy$$

$$\frac{1}{4} \int_0^2 \frac{u \cdot x du}{x} \int_0^2 \frac{u \cdot y du}{y}$$

$$= \frac{1}{4} \int_0^2 u du \int_0^2 u du$$

$$= \frac{1}{4} \left(\frac{u^2}{2} \right) \Big|_0^2 \left(\frac{u^2}{2} \right) \Big|_0^2$$

upper: $\ln e^2 = 2$, lower: $\ln(1) = 0$

$$U = \ln x \quad du = \frac{1}{x} dx \quad x du = dx$$

$$U = \ln y \quad du = \frac{1}{y} dy \quad y du = dy$$

$$= \frac{1}{4} \left(\frac{2}{2} - 0 \right) \left(\frac{2}{2} - 0 \right) \\ = \frac{1}{4} (2)(2) = \boxed{1} \rightarrow \text{one}$$

geometric
finite series, $r = 1/2$

$$\star 2. P_K(K) = \frac{1}{2^K} ; K = 1, 2, 3$$

$\star$

$$a) \sum_{i=1}^K \frac{1}{2^i} = 1 - \frac{1}{2^K} = \frac{\left(1 - \left(\frac{1}{2}\right)^K\right)}{1 - \frac{1}{2}} = 1 - \frac{1}{2^K}$$

$$F_K(K) = \begin{cases} 0 & K < 1 \rightarrow \text{zero otherwise} \\ 1 - \frac{1}{2^K} & K \geq 1 \rightarrow \text{for } K = 1, 2, 3 \dots \end{cases}$$

$\rightarrow$ known infinite series

$$b) E[2K-1]$$

$$E[K] = \sum_{K=1}^{\infty} K \cdot P_K(K)$$

$$= \sum_{K=1}^{\infty} \frac{K}{2^K} \rightarrow \frac{1/2}{\left(1 - \frac{1}{2}\right)^2} = \frac{\frac{1}{2}}{\frac{1}{4}} = 2$$

$$E[2K-1] = 2E[K] - 1$$

$$= 2 \cdot 2 - 1 = \boxed{3}$$

HW 4 con.

$$3b) \text{Var}(x) = E[x^2] - E([x])^2$$

$$E[x] = \int_a^b x f(x) dx + \int_c^d x f(x) dx$$

$$E[x] = \int_a^b x \cdot \frac{1}{(b-a)+(d-c)} dx + \int_c^d x \cdot \frac{1}{(b-a)+(d-c)} dx$$

$$= \frac{x^2}{2[(b-a)+(d-c)]} \Big|_a^b + \frac{x^2}{2[(b-a)+(d-c)]} \Big|_c^d$$

$$= \frac{b^2 - a^2}{2[(b-a)+(d-c)]} + \frac{d^2 - c^2}{2[(b-a)+(d-c)]}$$

$$= \frac{2[(b-a) + (d-c)]}{2[(b-a) + (d-c)]} \cdot \frac{(b^2 - a^2) + (d^2 - c^2)}{2[(b-a) + (d-c)]}$$

$$\begin{aligned} E[X^2] &= \int_a^b \frac{x^2}{(b-a) + (d-c)} dx + \int_c^d \frac{x^2}{(b-a) + (d-c)} dx \\ &= \frac{x^3}{3[(b-a) + (d-c)]} \Big|_a^b + \frac{x^3}{3[(b-a) + (d-c)]} \Big|_c^d \\ &= \frac{(b^3 - a^3) + (d^3 - c^3)}{3[(b-a) + (d-c)]} \end{aligned}$$

$$\text{Var}(X) = \frac{(b^3 - a^3) + (d^3 - c^3)}{3[(b-a) + (d-c)]} - \left(\frac{(b^2 - a^2) + (d^2 - c^2)}{2[(b-a) + (d-c)]} \right)^2$$



HW 4 con.

$$3. \text{ Pdf: } f(x) = \begin{cases} \frac{1}{(b-a) + (d-c)} & x \in [a, b] \cup [c, d] \\ 0 & \text{otherwise} \end{cases}$$

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt$$

$$\star a) \int_{-\infty}^{\infty} f_X(x) dx = \int_a^b \frac{1}{(b-a) + (d-c)} dx + \int_c^d \frac{1}{(b-a) + (d-c)} dx = 1$$

$$x < a: F_X(x) = 0$$

$$a \leq x < b: F_X(x) = \int_a^x \frac{1}{(b-a) + (d-c)} dt = \frac{x-a}{(b-a) + (d-c)}$$

$$\int_a^b \frac{1}{(b-a) + (d-c)} dx$$

$$b \leq x < c: F_x(x) = \frac{b-a}{(b-a)+(d-c)} \rightarrow \text{All probability mass from } a \text{ to } b \text{ is already accumulated}$$

$$c \leq x < d: F_x(x) = \frac{b-a}{(b-a)+(d-c)} + \int_c^x \frac{1}{(b-a)+(d-c)} dt$$

$$\rightarrow \frac{b-a}{(b-a)+(d-c)} + \frac{x-c}{(b-a)+(d-c)}$$

$$x \geq d: F_x(x) = 1$$

$$F_x(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{(b-a)+(d-c)} & a \leq x < b \\ \frac{b-a}{(b-a)+(d-c)} & b \leq x < c \\ \frac{b-a}{(b-a)+(d-c)} + \frac{x-c}{(b-a)+(d-c)} & c \leq x < d \\ 1 & x \geq d \end{cases}$$

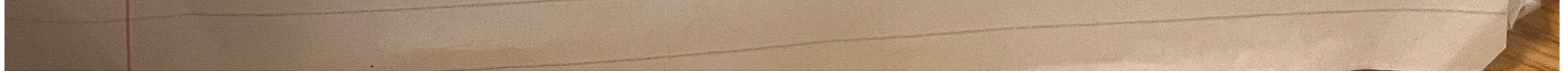
$$x < a$$

$$a \leq x < b$$

$$b \leq x < c$$

$$c \leq x < d$$

$$x \geq d$$



$\uparrow$ 2 (coin) + 2 scenarios $\rightarrow$ LoTP
 $\uparrow$ P (number of heads)

$C_1 = \text{coin one}$

$C_2 = \text{coin two}$

$$4a) P(X=x) = P(X=x|C_1) P(C_1) + P(X=x|C_2) P(C_2)$$

$$\hookrightarrow P(X=x) = \frac{1}{2} P(X=x|C_1) + \frac{1}{2} P(X=x|C_2)$$

$$\hookrightarrow \frac{1}{2} \binom{n}{x} p_1^x (1-p_1)^{n-x} + \frac{1}{2} \binom{n}{x} p_2^x (1-p_2)^{n-x}$$

• Factored out $\frac{1}{2} \binom{n}{x}$

$$\hookrightarrow \frac{1}{2} \binom{n}{x} \left(p_1^x (1-p_1)^{n-x} + p_2^x (1-p_2)^{n-x} \right)$$

$\uparrow$ combine like-terms

$$b) p_1 = p_2 = p$$

$$\hookrightarrow P(X=x) = \frac{1}{2} \binom{n}{x} p^x (1-p)^{n-x} + \frac{1}{2} \binom{n}{x} p^x (1-p)^{n-x}$$

$$\hookrightarrow \boxed{P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}} \rightarrow \boxed{X \sim \text{Bin}(n, p)}$$

Follows a Binomial(n, p) distribution

n = Number of cookie types sampled with prob. p

X = Number of Cookie types bought with prob. q

• Distribution of Y : $Y \sim \text{Binomial}(n, p)$ $\rightarrow$ satisfies conditions

PMF:

$$P(Y=K) = \binom{n}{K} p^K (1-p)^{n-K}$$

$\rightarrow$ Success probability p

$K = 0, 1, \dots, n$, n independent trials (cookie type)

• She must choose K from n different cookie types to sample

• Prob. of buying depends on if its sampled

• Distribution of X : $X \sim \text{Binomial}(n, pq)$

$\rightarrow$ Conditional

$\rightarrow$ Cookie type must first be sampled to be bought.

$\rightarrow$ Law of total prob.

$$\rightarrow P(\text{Bought}) = P(\text{Bought} | \text{Sampled}) \cdot P(\text{Sampled}) \rightarrow q \cdot p \rightarrow pq$$

PMF:

$$P(X=m) = \binom{n}{m} (pq)^m (1-pq)^{n-m}$$

$m = 0, 1, \dots, n$, n independent trials

• Can express X in terms of Y : $P(X=m) = \sum_{K=m}^n P(X=m | Y=K) P(Y=K)$

$$P(X=m) = \sum_{K=m}^n \binom{K}{m} q^m (1-q)^{K-m} \cdot \binom{n}{K} p^K (1-p)^{n-K} = \binom{n}{m} (pq)^m (1-pq)^{n-m} \quad \checkmark$$

$\rightarrow$ Law of total probability

$\rightarrow$ Goes to eqn. below

↑ All events independent

$\binom{n}{k} \rightarrow$ (choosing) which days result in pictures

6-3
A • No picture
either when she stays home or no one answers the door.

Hu 4 Con.

6. Probability she stays home: $1-p$, ($0 < p < 1$)

- Probability she visits the i th house: p
- Probability someone answers the door and she gives out pic: q ($0 < q < 1$)
- Probability no one answers and she goes home: $1-q$

6-1: $P(\text{Give out picture on 1st day}) = p \cdot q = \boxed{pq}$

↳ She has to go out (p), and someone answers door (q)

↳ Independent so we can multiply p and q

also
LOTP

$\binom{n}{k} \quad \binom{n}{n-k}$

can use $\uparrow$ 6-2: $P(\text{Exactly } k \text{ Pictures in } n \text{ days}) = \binom{n}{k} (pq)^k (1-pq)^{n-k}$

$X \sim \text{Binomial}(n, pq)$, $k=0, 1, 2, \dots, n$, n independent trials (days)
 n days, success prob. pq , each day is an independent bernoulli trial

If she stays home, then she has no chance to take a picture anyway.
 (Redundant)

(subset)

6-3: $P(\text{stayed home} \mid \text{No picture}) = \frac{P(\text{stayed home} \cap \text{No picture})}{P(\text{No picture})} = \frac{1-p}{1-pq}$
 $\nearrow$ same as $P(\text{stayed home})$

$P(1-q) \rightarrow$ Goes out and no one answers. Can multiply due to independence! $\rightarrow$ failure

$$P(\text{No picture}) = 1-p + p(1-q) = 1-pq$$

$\nearrow$ Complement of failures for 7 days

6-4: No days in which she went out and no one answered.

$\hookrightarrow$ She either stays home $(1-p)$, or goes out and gives a picture (pq) .

$$P(\text{No failures in 7 days}) = (1-p(1-q))^7$$

$$P(\text{No failures in a day}) = 1-p + pq = 1-p(1-q)$$

